MISB MOTION IMAGERY STANDARDS BOARD	MISB ST 1905.3
STANDARD **Motion Imagery Metadata (MIMD): Platform**	**2025-06-11**

1 Scope

The Motion Imagery Metadata (MIMD) Model [1] includes information about platforms that host payloads. The standard defines the modelling information about the different types of platforms and their specific attributes. The platform model includes classes for the Staging system and Payloads; for more information, please refer to MISB ST 1906 [2] and MISB ST 1907 [3] respectively.

2 References

[1] MISB, *ST 1903.3 - Motion Imagery Metadata (MIMD): Model,* 2025-06-11.

[2] MISB, *ST 1906.3 - Motion Imagery Metadata (MIMD): Staging System,* 2025-06-11.

[3] MISB, *ST 1907.3 - Motion Imagery Metadata (MIMD): Payload,* 2025-06-11.

[4] MISB, *ST 1901.3 - Motion Imagery Metadata (MIMD): Modeling Rules,* 2025-06-11.

[5] MISB, *MISP-2025.1: Motion Imagery Handbook,* 2025-06-11.

[6] MISB, *ST 1904.3 - Motion Imagery Metadata (MIMD): Base Attributes,* 2025-06-11.

[7] Federal Aviation Administration, *FAA-H-8083-25C - Pilot's Handbook of Aeronautical Knowledge,* 2023.

[8] World Meteorological Organization, *WMO-No. 8 - Guide to Instruments and Methods of Observation, Volume 1 - Measurement of Meteorological Variables, 2024 Edition.* (https://doi.org/10.59327/WMO/CIMO/1)

[9] ASM, *NGA.STND.0065_1.0 - National System for Geospatial Intelligence Taxonomy (NTAX) Standard,* 2017-04-11.

3 Revision History

Version	Approved Date	Change Description
ST 1905.3	2025-06-11	• Deprecated Platform class's name and identity attributes; both attributes are part of the Device Identity class, via the deviceId attribute

4 Definitions of Terms, Acronyms, and Initialisms

4.1 Terms

The following lists the terms and their definitions this document uses.

Platforms	Devices which host payloads (e.g., sensors, etc.); for example ships, planes, tanks

4.2 Acronyms and Initialisms

The following lists the acronyms and initialisms this document uses.

AGL	Above Ground Level
ISR	Intelligence, Surveillance & Reconnaissance
MIMD	Motion Imagery Metadata
MISB	Motion Imagery Standards Board
MISP	Motion Imagery Standards Profile
NGA	National Geospatial-Intelligence Agency
SI	International System of Units
ST	Standard
UML	Unified Modelling Language

5 Introduction

Platforms host payloads including Motion Imagery sensors. There are a wide variety of platforms with different requirements (e.g., traverse land, sea, or fly) and performing different functions (e.g., tracking, ISR) in different environments (e.g., air, water, high altitude). The simplest platform to describe is a handheld camera; the "platform" is the person without a known position or orientation. An example complex platform is a remote-controlled or automated system with multiple payloads and weapons in a hostile environment. With a remote-controlled platform, operators need information about the environment, the propulsion unit status, on-board storage and more. Analysts use this information to fully understand the situation when collecting the Motion Imagery.

Different types of platforms need and produce different types of information but also share similar types of information. For example, airborne platforms need different types of altitude (e.g., AGL) while a surface platform does not; however, both platform types need position and orientation information.

The MIMD Model for platforms provides attributes and classes to support the wide range of platform types and environments.

6 Platforms Overview

Platforms host one or more payloads and likely include power systems and weapons. Platforms operate in an environment, such as in the air, on the ground and water surface, or under water. The platform may transition from one environment to another during its operation.

Platform characteristics include administrative, position, orientation, locomotion, environment, weapons, payloads, and other devices. The administrative data provides identity, operational information, and status. The position data describes where the platform is located on or around the earth and includes position kinematics. The orientation data describes the direction the platform is facing in all three dimensions and includes orientation kinematics. The locomotion data describes the method and status of the powerplant for moving and orientating the platform (e.g., power levels, fuel levels). Environmental data describes the medium(s) and their status (e.g., air temperatures, water temperatures, air pressure). Weapons data describes weapons and their status on the platform. The payloads data describes the on-board sensors, their status, and relevant information for sensor data exploitation. Other devices include on-board storage, transmitters, receivers, encoders, etc. Some of these characteristics are future additions to the Platform class.

7 Model Classes

7.1 Platform Class

Figure 1 shows the UML class diagram for the MIMD Platform. The Platform class provides administrative attributes, along with a Stage class list (see MISB ST 1906), Payload class list (see MISB ST 1907), Atmosphere class, and SurfaceWater class.

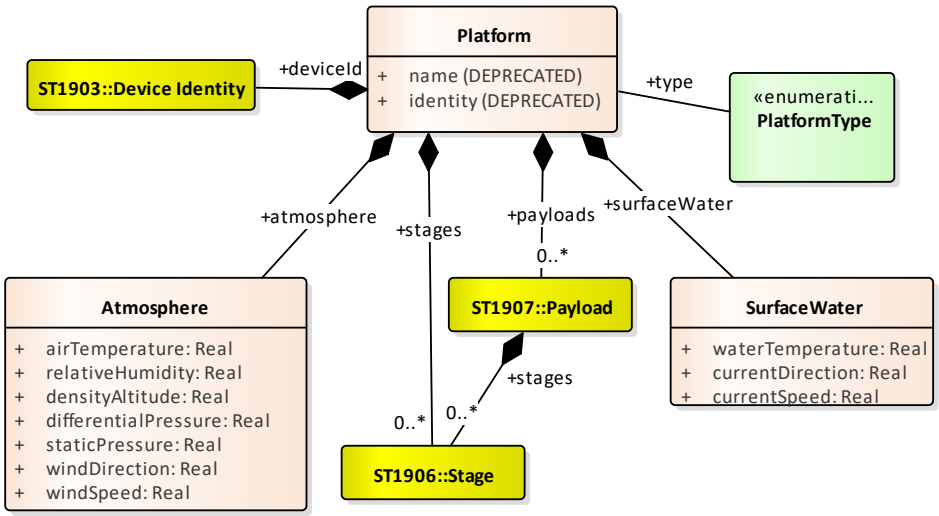


Figure 1: Platform Class Model

The Atmosphere class provides attributes about the atmospheric conditions along with the orientation of the platform for the given conditions. The SurfaceWater class provides attributes about the water conditions along with the orientation of the platform for the given conditions. Platforms, such as a ship, may report both Atmosphere and SurfaceWater conditions in the same MIMD packets.

Per discussion in MISB ST 1901 [4], MIMD classes support the Report-On-Change (reference the Motion Imagery Handbook [5]) and include MIMD Base Attributes (reference MISB ST 1904 [6]), which use attribute identifiers 1-32.

Table 1 lists the attributes for the Platform class.

Table 1: Platform Class Attributes[1]

Id	Name	Type	Min	Max	MLen	Units	Ref
~~33~~	~~name~~ **(DEPRECATED)**	~~String~~	~~N/A~~	~~N/A~~	~~100~~	~~None~~	~~7.1.1~~
~~34~~	~~identity~~ **(DEPRECATED)**	~~String~~	~~N/A~~	~~N/A~~	~~100~~	~~None~~	~~7.1.2~~
35	type	PlatformType	N/A	N/A	N/A	None	7.1.3
36	atmosphere	Atmosphere	N/A	N/A	N/A	None	7.1.4
37	surfaceWater	SurfaceWater	N/A	N/A	N/A	None	7.1.5
38	stages	LIST<Stage>	--	--	N/A	None	7.1.6
39	payloads	LIST<Payload>	--	--	N/A	None	7.1.7
40	deviceId	DeviceIdentity	N/A	N/A	N/A	None	7.1.8

7.1.1 ~~Attribute 33 – name~~ (DEPRECATED)

The *name* attribute of the DeviceIdenty class replaces the Platform class's *name* attribute; see the Platform's deviceId attribute.

7.1.2 ~~Attribute 34 – identity~~ (DEPRECATED)

The *identity* attribute of the DeviceIdenty class replaces the Platform class's *identity* attribute; see the Platform's deviceId attribute.

7.1.3 Attribute 35 – type

This attribute is an enumeration. Section 8.1 defines the PlatformType enumeration.

The *type* attribute specifies the kind of platform from the PlatformTypes enumeration.

7.1.4 Attribute 36 – atmosphere

This attribute is a composite relationship to the Class Atmosphere. Section 7.2 defines the Atmosphere class.

7.1.5 Attribute 37 – surfaceWater

This attribute is a composite relationship to the Class SurfaceWater. Section 7.3 defines the SurfaceWater class.

[1] For consistency across the MIMD suite of documents, MISB ST 1901 defines the label designations for the columns in the attribute tables.

7.1.6 Attribute 38 – stages

This attribute is a list of Stage classes. MISB ST 1906 defines the Stage class.

Platforms that move in a nominal primary direction (e.g., ships and planes but not quadcoptors) define their forward direction down the x-axis, the y-axis on the left-hand side, and z in the upward direction. Non-directional platforms may choose any sensible orientation, so that the z-axis points in the upwards direction.

MISB ST 1906 defines the Stage class and all its child classes.

7.1.7 Attribute 39 – payloads

This attribute is a list of Payload classes. MISB ST 1907 defines the Payload class.

7.1.8 Attribute 40 – deviceId

This attribute is a composite relationship to the Class DeviceIdentity. MISB ST 1903 defines the DeviceIdentity class.

The DeviceIdentity *name* attribute specifies a label for the platform. The name is a word or set of words for which the platform is known. For example, "Enterprise" is the name of a U.S. Navy aircraft carrier.

The DeviceIdentity *identity* attribute specifies an organization's identifier for the platform. The identity is the organization's code, number, word, or set of words, the organization uses to identify the platform. The identity should be unique to the platform. For example, "Enterprise" is the name of a U.S. Navy aircraft carrier; however, there are several ships with that same name. Using the identifier "CVN-65", uniquely identifies a specific ship.

7.2 Atmosphere Class

The Atmosphere class defines attributes for air conditions external to the platform. Table 2 lists the attributes for the Atmosphere class.

Table 2: Atmosphere Class Attributes[2]

Id	Name	Type	Min	Max	MLen	Units	Ref
33	airTemperature	Real	-274.0	--	N/A	°C	7.2.1
34	relativeHumidity	Real	0.0	100.0	N/A	%	7.2.2
35	densityAltitude	Real	-12 000.0	--	N/A	m	7.2.3
36	differentialPressure	Real	0.0	--	N/A	hPa	7.2.4
37	staticPressure	Real	0.0	--	N/A	hPa	7.2.5
38	windDirection	Real	0.0	TWO_PI	N/A	rad	7.2.6
39	windSpeed	Real	0.0	--	N/A	m/s	7.2.7

[2] For consistency across the MIMD suite of documents, MISB ST 1901 defines the label designations for the columns in the attribute tables. Capitalized attribute table items (e.g., PI) are constant values listed in MISB ST 1901.

7.2.1 Attribute 33 – airTemperature

The airTemperature attribute is the measured ambient air temperature outside of the platform. The minimum temperature is absolute zero.

7.2.2 Attribute 34 – relativeHumidity

The relativeHumidity attribute is the measured ambient humidity outside of the platform. Relative Humidity is the ratio between the water vapor density and the saturation point of water vapor density expressed as a percentage.

7.2.3 Attribute 35 – densityAltitude

The densityAltitude attribute is the density altitude as defined by the FAA Pilot's Handbook of Aeronautical Knowledge [7]. The Density Altitude is the vertical distance above sea level in the standard atmosphere at which a given density is found. Density Altitude is pressure altitude corrected for nonstandard temperature. As the density of the air increases (lower density altitude), aircraft performance increases; conversely as air density decreases (higher density altitude), aircraft performance decreases.

7.2.4 Attribute 36 – differentialPressure

The differentialPressure attribute is the difference between the impact pressure the platform is experiencing (e.g., a measure from an aircraft's pitot tube) and the static pressure outside of the platform. Differential pressure provides a method of calculating relative velocity of an item as it passes through a fluid, or conversely the velocity of a fluid as it passes by an item. Velocity can be determined by differential pressure by the following:

$$v_1 = \sqrt{\frac{2p_d}{\rho}}$$

where p_d is the measured differential pressure (p_d = impact pressure minus static pressure = $p_i - p_s$), and ρ is the density of the fluid outside the item.

The SI unit for pressure is Pascals. However, per the World Meteorological Organization's (WMO) Guide to Meteorological Instruments and Methods of Observation [8], when reporting atmospheric pressure, the preferred SI prefix is hecto; therefore, atmospheric pressure measures in the MIMD Model are in hectopascals (hPa). The WMO has chosen hPa as the preferred measure since 1 millibar = 1 hPa, where the millibar is the legacy unit for measuring atmospheric pressure.

7.2.5 Attribute 37 – staticPressure

The staticPressure attribute is the pressure of the air that surrounds the platform. Platforms measure static pressure by a sensor mounted outside of the platform and out of the air stream. Static pressure, along with impact pressure provides the means to compute indicated airspeed, true airspeed, and density altitude.

See attribute 36 for information about the use of the hectoPascals (hPa) units.

7.2.6 Attribute 38 – windDirection

The windDirection attribute is a measure of what direction the air body, or wind, (around the platform) is moving, relative to true north.

7.2.7 Attribute 39 – windSpeed

The windSpeed attribute is a measure of the velocity of the air body, or wind, relative to the ground or earth.

7.3 SurfaceWater Class

The SurfaceWater class defines attributes for water conditions external to the platform when the platform is on the surface of water. Table 3 lists the attributes for the SurfaceWater class.

Table 3: SurfaceWater Class Attributes[3]

Id	Name	Type	Min	Max	MLen	Units	Ref
33	waterTemperature	Real	-274.0	--	N/A	°C	7.3.1
34	currentDirection	Real	0.0	TWO_PI	N/A	rad	7.3.2
35	currentSpeed	Real	0.0	--	N/A	m/s	7.3.3

7.3.1 Attribute 33 – waterTemperature

The *waterTemperature* attribute is the temperature measurement of the water at or near the surface next to the platform. The minimum temperature is absolute zero.

7.3.2 Attribute 34 – currentDirection

The *currentDirection* attribute is the angle relative to true north the water current is flowing from the present platform position outward.

7.3.3 Attribute 35 – currentSpeed

The *currentSpeed* attribute is the magnitude of the current's velocity.

[3] For consistency across the MIMD suite of documents, MISB ST 1901 defines the label designations for the columns in the attribute tables. Capitalized attribute table items (e.g., PI) are constant values listed in MISB ST 1901.

8 Model Enumerations

8.1 PlatformType Enumeration

The NSG Taxonomy [9] (NTAX) provides many of the items in the PlatformType Enumeration; please refer to this reference for each item's detailed description. The Enumeration contains general items (e.g., Aircraft) and specific items within the general item's category (e.g., FixedWingAircraft). Producers should use the most specific item that best describes their platform.

Table 4: Enumeration Values for PlatformType

Id	Name	Description
1	Aircraft	A vehicle designed primarily for transportation in air within the atmosphere
2	FixedWingAircraft	An aircraft capable of flight using fixed wings that generate lift caused by the forward airspeed of the vehicle and the shape of its wings
3	FixedWingMannedAircraft	A manned aircraft capable of flight using fixed wings that generate lift caused by the forward airspeed of the vehicle and the shape of its wings
4	FixedWingUAV	An unmanned aircraft capable of flight using fixed wings that generate lift caused by the forward airspeed of the vehicle and the shape of its wings
5	RotaryWingAircraft	An aircraft that is an aerodyne having wings that rotate about an axis, usually an approximately vertical axis
6	RotaryWingMannedAircraft	A manned aircraft that is an aerodyne having wings that rotate about an axis, usually an approximately vertical axis
7	RotaryWingUAV	An unmanned aircraft that is an aerodyne having wings that rotate about an axis, usually an approximately vertical axis
8	GroundVehicle	A vehicle designed primarily for transportation upon land
9	ArmouredCombatVehicle	A self-propelled, armoured ground vehicle designed for cross-country mobility and equipped with an integral or organic weapon
10	ArmouredGroundPlatform	A standardized armoured ground vehicle that consists of an integrated engine, transmission, driveshaft, suspension, and tracks, enclosed in an armoured hull to which is mounted alternative mission-specific systems (for example: anti-tank weapon, ammunition resupply equipment, field artillery, engineering equipment)
11	ArmouredRecoveryVehicle	An armoured ground vehicle designed to recover other armoured vehicles
12	HalfTrackedArmouredVehicle	An armoured ground vehicle with motive power provided by linked traction belts and steering accomplished by front wheels and/or rear tracks
13	HeavyArmouredCombatVehicle	A combat ground vehicle that combines firepower with thick armour, thereby sacrificing some speed and mobility in favor of superior defensive protection
14	LightArmouredCombatVehicle	A combat ground vehicle that combines firepower with increased speed and mobility due to its lighter armour protection
15	LightTank	A tracked combat vehicle, equipped with lightweight artillery and protected by thin armour, that is designed to perform combat duties (for example: reconnaissance, amphibious operations, support to airborne divisions) requiring speed, maneuverability, and mobility
16	MainBattleTank	A heavily armoured, tracked combat vehicle with high cross-country mobility and capable of heavy firepower from a large direct-fire main gun

17	ArtilleryCommandVehicle	A ground vehicle designed to transport and/or provide workspace where the artillery unit commander or staff fire-control element can perform its artillery-associated tasks
18	EngineeringVehicle	A ground vehicle designed specifically for military missions and/or civil engineering, having the potential for military engineering applications in a combat theatre of operations
19	ArmouredBridgeLauncher	A large, fully armoured engineering vehicle capable of transporting and launching a prefabricated mechanical bridge
20	CraneVehicle	An engineering vehicle with an integrated machine (typically consisting of a boom, hoist, wire rope or chain, pulleys, and hook) designed to raise or lower objects, and to move them horizontally within its rotational radius
21	EarthMovingEquipment	An engineering vehicle designed to disrupt, move and/or transport natural or man-made ground materials in support of civil engineering activities
22	Bulldozer	A wheeled or tracked engineering vehicle with a horizontal blade used for clearing or grading
23	Grader	A wheeled engineering vehicle having a horizontal blade and designed for use in levelling operations in horizontal construction
24	MobileExcavator	A heavy construction engineering vehicle designed to excavate and/or transport loose material from the ground by means of an articulated or boomed bucket
25	MultipurposeEarthMover	A tracked and wheeled engineering vehicle optionally outfitted with various pieces of engineering devices (for example: dozer blades, backhoes) and often fitted with more than one such device
26	Scraper	A large engineering vehicle designed with a vertically moveable hopper (also known as the bowl) that has a sharp horizontal front edge capable of cutting horizontally into soil or clay in order to fill the hopper
27	Trencher	A wheeled or tracked engineering vehicle used to dig ditches or trenches
28	GroundAmbulance	A ground vehicle designed and equipped for transporting wounded, sick, or injured personnel
29	ArmouredTrackedAmbulance	A ground ambulance equipped with linked traction belts, armoured for defense purposes, and designed for transporting wounded, sick, or injured personnel in battlefield conditions
30	ArmouredWheeledAmbulance	A ground ambulance equipped with wheels for mobility, armoured for defense purposes, and designed for transporting wounded, sick, or injured personnel in battlefield conditions
31	TruckWheeledAmbulance	A ground ambulance based on an un-armoured or very lightly armoured, wheeled, truck-style vehicle, and used for transporting wounded, sick, or injured personnel
32	PetroleumOilLubVehicle	A ground vehicle used to provide transportation and storage of petroleum, oil, and lubricants
33	PetroleumOilLubPipeLayer	A vehicle designed and used for laying and recovering tactical fuel hose lines and pipelines

34	PetroleumOilLubTanker	A vehicle designed to transport and dispense fuel for ground vehicles, aircraft, and equipment
35	UnmannedGroundVehicle	A ground vehicle without a human driver aboard guided through its tasks either remotely or autonomously (i.e., by an internal computer program)
36	WheeledCargoTransport	A wheeled ground vehicle designed for transporting loads on- and/or off-road
37	CargoTruck	A ground vehicle specially designed to carry large and/or heavy loads in a cargo bed
38	DumpTruck	A ground vehicle equipped with an open-box bed, which is hinged at the rear and equipped with hydraulic pistons to lift the front, allowing the material in the bed to be deposited
39	SemiTrailerTruck	An automotive ground vehicle consisting of a towing engine (for example: truck tractor) with an attached semi-trailer used to carry freight
40	Trailer	A non-self-powered ground vehicle designed to carry cargo and to be towed by another vehicle
41	TruckTractor	A ground vehicle designed to support and tow a semi-trailer, which is attached using a swiveling lock (usually a fifth wheel coupling) mounted on the truck tractor
42	UtilityTruck	An automotive ground vehicle constructed with a small heavy rectangular frame that includes an integrated rear cargo area of limited load-carrying and volumetric capacity
43	MaritimeVessel	A vehicle designed for travel in or under water that combines buoyancy, internal capacity (tonnage), seaworthiness (stability and handling), propulsion, and (if self-powered) speed in an appropriate configuration for its intended purpose or mission
44	AmphibiousTransporter	A maritime vessel designed to move personnel and equipment across wetlands or inland bodies of water
45	BridgeSupportBoat	A maritime vessel designed primarily to support construction and maintenance of bridges but having sufficient power to transport equipment or personnel directly across the water (typically on rafts)
46	Ferry	A maritime vessel designed for transporting passengers, vehicles, and/or goods across a stretch of water
47	Hulk	A vessel, either stranded aground or permanently moored, that is no longer seaworthy due to an inoperable propulsion plant or compromised vessel integrity
48	NavalAuxiliaryShip	A maritime vessel designed to provide logistical support, including replenishment and repair, to another ship while in the open ocean, also capable of use in port
49	NavalCombatantCraft	A smaller maritime vessel that has limited operational endurance, utilized primarily in coastal areas to apply force in one or more types of missions
50	NavalCombatantShip	A maritime vessel designed to be involved directly in the application of military power in the open ocean, for one or more types of warfare missions

51	NavalSubmarine	A self-propelled submersible maritime vessel with at least a residual combat capability
52	NavalSupportCraft	A maritime vessel designed primarily for harbor or base-support duties in a non-combatant role
53	Pontoon	An airtight air-filled hollow structure, usually rectangular in shape, which is buoyant in water
54	Spacecraft	A vehicle designed to travel in outer space
55	Satellite	A spacecraft designed to perform a specific function when placed in orbit around the Earth or another body in outer space
56	SpaceLaunchVehicle	A spacecraft consisting of a rocket designed to carry a payload from Earth's surface into outer space.
57	Automobile	Platform is a four-wheeled commercial vehicle for carrying people (e.g., sedan, SUV, hatchback)
58	MotorCycle	Platform is a two or three wheeled motorized bike
59	Cycle	Platform is a two or three wheeled human powered bike
60	Animal	Platform is a living creature for body mounted sensor(s)
61	Human	Platform is a person for handheld or body mounted sensors(s)
62	Structure	Platform is a building or other structure for mounting sensor(s)
63	Pole	Platform is a pole (e.g., telephone pole or sensor tower), statically mounted in the ground
64	OpticalTrackingMount	Platform is a self-contained gimbal device for pointing sensors